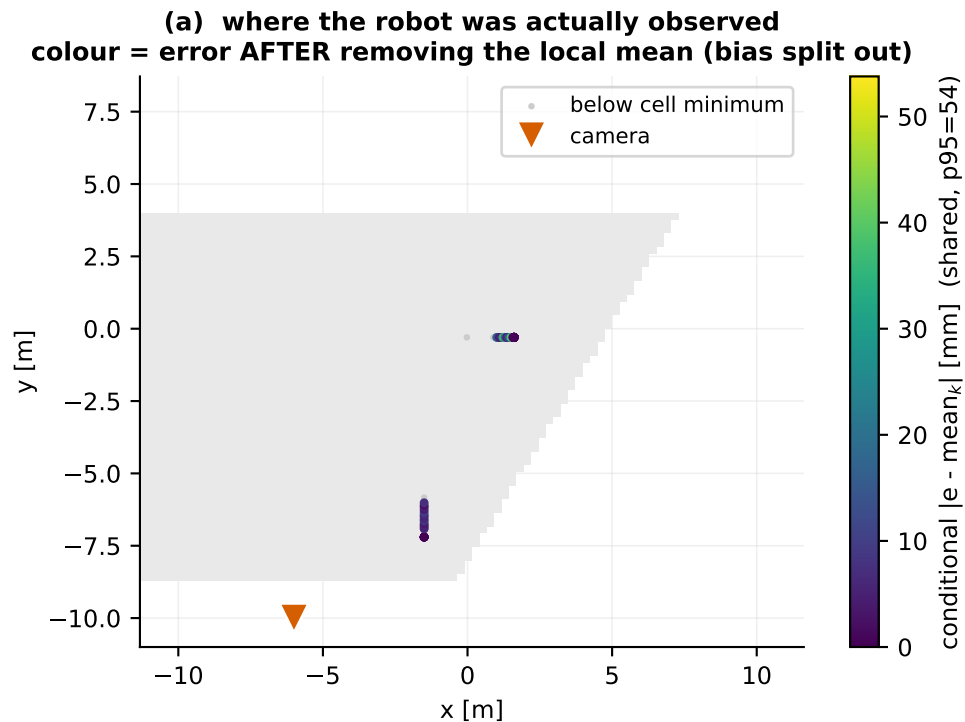
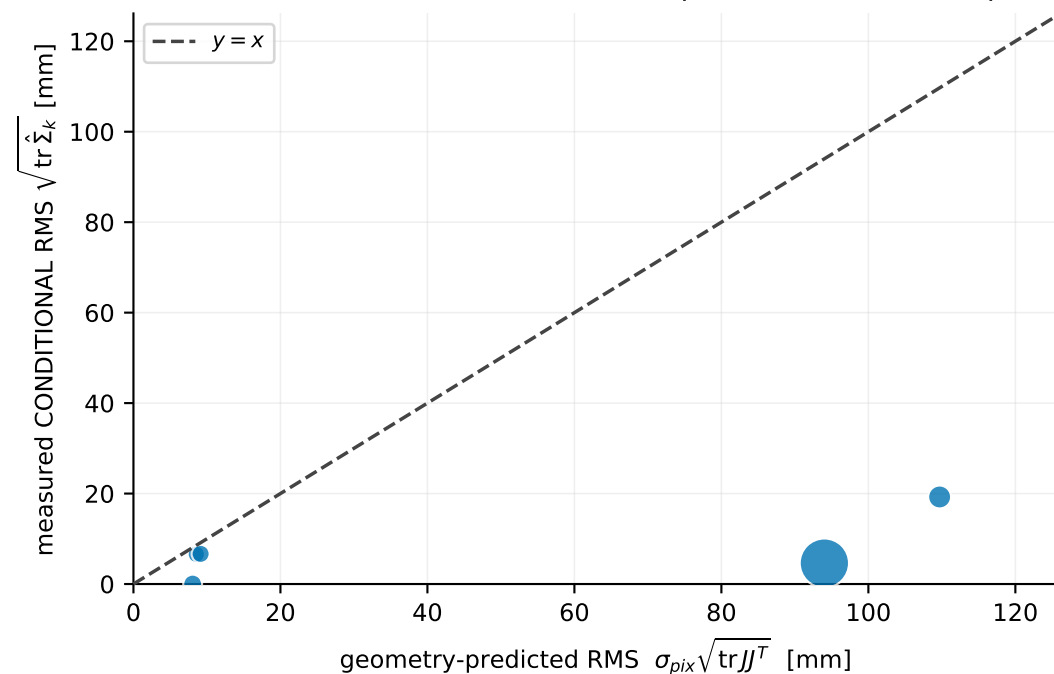


Camera A — deployed-corrected residual



(b) does geometry predict the local SPREAD?
N = 125 detections · M = 5 cells (0.5 m, ≥ 5 each)
conditional: R^2 vs identity = -78.62 · cell Spearman = +0.70 · median obs/pred = 0.18
median per-cell BIAS removed first = 80 mm
ONLY 5 CELLS — too few to establish a trend; read as a scale check, not a fit



Point area in (b) scales with detections in the cell. σ_{pix} is fitted on leave-region-out folds, so each cell's prediction is out-of-region. Ground truth positions the dots in (a) and measures the residual; it never enters a projection, Jacobian or covariance. **READ THE LIMIT BEFORE THE RESULT:** the sampled ribbon in (a) never covers the footprint in two dimensions, so range, image row and projection conditioning are collinear along it. A weak or negative R^2 here therefore does NOT show that geometry fails -- it shows this dataset cannot test it. The honest conclusion is that straight-route logs are insufficient to decide whether full 2-D projection geometry beats a simpler range effect; that needs a commissioning capture with 2-D coverage and multiple headings per location.